Lesson 12: Hearing It Happen

Godot Engine

VIDEO GAME CREATION

This week: your game finally sounds like a game. Four moments - firing, hitting, clearing a wave, and losing - each get their own distinct cue.

Open your own [YourName]Game project from Lesson 11 - don't start a new one.

If you already built an optional sound idea in an earlier lesson (Lessons 2, 3, 4, or 6), keep it - today reuses it directly instead of replacing it. If you didn't, today's steps are written as if starting fresh.

Setup

1. Open your [YourName]Game project from Lesson 11.
2. Add a new AudioStreamPlayer2D node to GameManager, named WaveSound - today's one genuinely new sound, alongside the WinSound/LifeLostSound/GameOverSound nodes Lessons 6 and 8 already built (if you skipped those, add GameOverSound now too - it becomes mandatory starting today).
3. Import sfx_wave.wav (Resources > Import > Sound) alongside your existing sfx_shoot.wav, sfx_hit.wav, and sfx_gameover.wav/sfx_lifelost.wav/sfx_win.wav.

Level 1 - Minimal (MVP)

Four sounds fire at the correct moment, at least once each: a shot, a hit/explosion, a wave-clear (on any non-final wave), and a game over. No silence, no obviously wrong triggers.

4. On the Player, confirm your AudioStreamPlayer2D.play() already fires in your fire function next to bullet instantiation (from Lesson 3) - add it now if it's missing.
5. On the Enemy scene's script, confirm its own AudioStreamPlayer2D plays sfx_hit.wav when destroyed (from Lesson 4) - add it now if it's missing.
6. Assign the new WaveSound node's Stream to sfx_wave.wav in the Inspector.
7. In on_enemy_destroyed()'s nested if (from Lesson 11), add $WaveSound.play() to the wave-advance (else) branch.
8. On GameManager, confirm $GameOverSound.play() already fires in on_enemy_escaped()'s lives-reaches-zero branch (from Lesson 8) - add the node and the line now if either is missing. This sound is mandatory starting today even if you skipped it before.
9. Run the scene and play through a full round - shoot, hit, an early wave clear, and a loss should each sound distinct.

Highlighted lines are new this week; everything else is code you already have.

```
func on_enemy_destroyed():
    enemies_remaining -= 1
    score += 10
    update_hud()
    if enemies_remaining <= 0:
        if wave >= total_waves:
            game_active = false
            get_node("../WinText").text = "You Win! Press R to Restart"
            get_node("../WinText").show()
            $WinSound.play()  # your Lesson 6 Level 3 sound
        else:
            wave += 1
            enemies_remaining = 5
            update_hud()
            $WaveSound.play()
            get_parent().call_deferred("spawn_enemies")

func on_enemy_escaped():
    lives -= 1
    update_hud()
    if lives <= 0:
        game_active = false
        get_node("../LoseText").text = "Game Over - Press R to Restart"
        get_node("../LoseText").show()
        $GameOverSound.play()  # your Lesson 8 Level 3 sound, mandatory as
of today
    else:
        $LifeLostSound.play()  # your Lesson 8 Level 3 sound
```

Level 2 - Complete (Alpha/Beta)

All four sounds fire reliably every time, with no missing or duplicate playback, and each is clearly distinguishable from the others by ear.

- Play a full round: shoot, hit, an early wave clear, and a loss each produce a distinct sound.
- If you built the optional win sound, confirm it still plays on the true win, and WaveSound doesn't also fire on that same frame.
- Confirm $WaveSound, like $WinSound/$LifeLostSound/$GameOverSound, is its own dedicated node - no shared stream-swapping needed, since each node only ever plays its own one clip.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Audio: if you haven't already, add the optional win sound now - it's the same $WinSound.play() pattern Lesson 6 built, alongside today's WaveSound.
- Audio: add slight pitch variation to hits by setting the Enemy's AudioStreamPlayer2D pitch_scale to a random value (e.g. randf_range(0.9, 1.1)) right before each .play() call.
- Structure: notice every round-ending sound (WinSound, WaveSound, LifeLostSound, GameOverSound) is its own small, dedicated node living on GameManager, right next to the state change it represents - four simple nodes instead of one more complicated shared one.

If Something's Wrong

- No sound on wave-clear: confirm $WaveSound.play() is actually inside the else branch, not after the if/else.
- NullReferenceException-style "Node not found" on WaveSound: confirm the node is actually named WaveSound (case-sensitive) and is a direct child of GameManager.
- Game-over or life-lost sound plays repeatedly: confirm the lose branch is still guarded by "if lives <= 0" / "else" as a one-time check (from Lesson 8), not re-triggering every frame.

Want More?

Student Resource List > Godot section, or docs.godotengine.org